

Matrix Algebra

Matrix: A system of any numbers arranged in a rectangular array of m rows and n columns is called a matrix of order $m \times n$. A matrix is usually denoted by a single capital letter, namely A, B, C, ... or by the symbols $[a_{ij}]$, (a_{ij}) , $\|a_{ij}\|$.

The matrix of order $m \times n$ is written as:

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n}$$

Example: $\mathbf{A} = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}_{3 \times 3}$; $\mathbf{B} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}_{3 \times 1}$; $\mathbf{C} = [1 \ 2 \ 3]_{1 \times 3}$; $\mathbf{D} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}_{2 \times 3}$.

Question: Distinguish between a matrix and a determinant.

Answer: The differences between a matrix and a determinant are as follows:

Matrix	Determinant
1. A matrix cannot be reduced to a single number.	1. A determinant can be reduced to a single number.
2. In a matrix, the number of rows may not be equal to the number of columns.	2. In a determinant, the number of rows must be equal to the number of columns.
3. An interchange of rows or columns gives a different matrix.	3. An interchange of rows or columns gives the same determinant with +ve or -ve sign.
4. Examples: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$; $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$.	4. Examples: $\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$; $\begin{vmatrix} 1 & 2 & 3 \\ -2 & 1 & 0 \\ 2 & 3 & 4 \end{vmatrix}$.

Question: Define rectangular and square matrix.

Answer: Rectangular Matrix: A matrix **A** of order $m \times n$ is called a rectangular matrix if the number of rows and the number of columns are not equal **i.e.**, $m \neq n$.

Example: $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 1 \end{bmatrix}; B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}.$

Square Matrix: A matrix A of order $m \times n$ is called a square matrix if the number of rows and the number of columns are equal **i.e.** $m = n$.

Example: $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}; B = \begin{bmatrix} 1 & -2 & 3 \\ 2 & 3 & -1 \\ -3 & 1 & 2 \end{bmatrix}$

Question: Define Row and Column matrix.

Answer: Row Matrix: A matrix A is called a row matrix row vector if it contains only one row.

Example: $A = [1 \ 2 \ 3].$

Column Matrix: A matrix A is called a column matrix or column vector if it contains only one column.

Example: $A = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}.$

Question: Define Null matrix and Identity matrix.

Answer: Null Matrix: A matrix, rectangular or square, each of whose elements is zero is called a **Zero matrix** or **Null matrix** and is denoted by **O**.

Example: $A = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}; B = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$

Identity Matrix: A square matrix whose elements $a_{ij} = 0$ when $i \neq j$ and $a_{ij} = 1$ when $i = j$ is called the **Identity matrix or Unit matrix** and is denoted by **I**.

Example: $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

Question: Define Diagonal matrix and Scalar matrix.

Answer: Diagonal Matrix: A square matrix whose elements $a_{ij} = 0$ when $i \neq j$ is called a **Diagonal matrix**. The elements a_{ij} when $i = j$ are known as diagonal elements and the line along which they lie is known as the **principal diagonal** or **leading diagonal**.

$$\text{Example: } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 3 \end{bmatrix}$$

Scalar Matrix: If in a square matrix **A** all the diagonal elements are equal to a nonzero element **a** and all the remaining elements are equal to zero then it is called a **Scalar matrix**.

$$\text{Example: } A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$$

Question: Define Triangular, Upper Triangular and Lower Triangular matrices.

Answer: Triangular Matrix: A square matrix whose elements $a_{ij} = 0$ when $i > j$ or $i < j$ is called a **Triangular matrix**.

$$\text{Example: } A = \begin{bmatrix} 1 & 3 & 2 & 4 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 5 & 7 \end{bmatrix}; B = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}$$

Upper Triangular Matrix: A square matrix **A** whose elements $a_{ij} = 0$ when $i > j$ is called an **upper triangular matrix**.

$$\text{Example: } A = \begin{bmatrix} 2 & 3 & 4 & 5 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 2 & 5 \\ 0 & 0 & 0 & 7 \end{bmatrix}$$

Lower Triangular Matrix: A square matrix **A** whose elements $a_{ij} = 0$ $i < j$ is called a **lower triangular matrix**.

Example: $A = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 3 & 1 & 0 & 0 \\ 4 & 3 & 2 & 0 \\ 5 & 4 & 5 & 7 \end{bmatrix}$

Question: Define sub-matrix of a matrix.

Answer: A matrix, which is obtained from a given matrix by deleting any number of rows and columns or both, is called a **sub-matrix** of the given matrix.

Example: If $A = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 5 & 6 & 7 & 8 \\ 0 & 7 & 5 & 2 \end{bmatrix}$ be a matrix then $\begin{bmatrix} 1 & 2 & 3 \\ 5 & 6 & 7 \\ 0 & 7 & 5 \end{bmatrix}$; $\begin{bmatrix} 3 & 4 \\ 7 & 8 \end{bmatrix}$ etc. are its sub-matrices.

Question: Define equality of matrices.

Answer: Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are said to equal iff

- They are of the same dimensions **i.e.**, they are of the same order.
- The elements in the corresponding positions of the two matrices are same.

Question: Define order or dimension of a matrix.

Answer: The **order** or **dimension** of a matrix is given by stating the number of rows and the number of columns in the matrix.

Example: $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 4 & 3 & -2 & 0 \\ -2 & 1 & 0 & 4 \end{bmatrix}$ is a matrix of order 3×4 .

Question: Explain the addition and subtraction of matrices.

Answer: Addition: If **A** and **B** be two matrices of order $m \times n$ given by $A = [a_{ij}]$ and $B = [b_{ij}]$, then the matrix $A + B$ is defined as the matrix each element of which is the sum of the corresponding elements of **A** and **B** i.e. $A + B = [a_{ij} + b_{ij}]$, where $i = 1, 2, 3, \dots, m$ and $j = 1, 2, 3, \dots, n$.

Example: If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ then $A + B = \begin{bmatrix} 1+2 & 2+3 \\ 3+4 & 4+5 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 7 & 9 \end{bmatrix}$.

Subtraction: If A and B be two matrices of order $m \times n$ given by $A = [a_{ij}]$ and $B = [b_{ij}]$, then the matrix $A - B$ is defined as the matrix each element of which is obtained by subtracting the elements of B from the corresponding elements of A i.e. $A - B = [a_{ij} - b_{ij}]$, where $i = 1, 2, 3, \dots, m$ and $j = 1, 2, 3, \dots, n$.

Example: If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ then $A - B = \begin{bmatrix} 1-2 & 2-3 \\ 3-4 & 4-5 \end{bmatrix} = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix}$.

Question: Define conformable for addition and subtraction.

Answer: If the two matrices A and B are of the same order, then only their addition and subtraction is possible and these matrices are said to be **Conformable** for **Addition** or **Subtraction**.

Question: Explain the multiplication of matrices.

Answer: If A and B be two matrices such that the number of columns in A is equal to the number rows in B i.e. if $A = [a_{ij}]$ and $B = [b_{jk}]$ are $m \times n$, $n \times p$ matrices then the product of the matrices A and B denoted by AB is defined as matrix

$$C = [c_{ik}]$$

$$= \sum_{j=1}^n a_{ij} b_{jk}$$

In the matrix product AB , the matrix A is called the pre-multiplier and B is called the post-multiplier.

Example: If $A = \begin{bmatrix} 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$ then $AB = \begin{bmatrix} 1+4 & 0+6 \end{bmatrix} = \begin{bmatrix} 5 & 6 \end{bmatrix}$.

Question: Define transpose of a matrix.

Answer: The matrix obtained from any given matrix A by interchanging its rows and columns is called its transpose. The transpose of A , is denoted by A^t or A' .

Example: If $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & -2 \end{bmatrix}$, then $A^t = \begin{bmatrix} 1 & 0 & 1 \\ 2 & 1 & -2 \end{bmatrix}$.

NOTE: The transpose of the product of two matrices is equal to the product of their transpose in reverse order, i.e. $(AB)^t = B^t A^t$.

Question: Define determinant of a square matrix.

Answer: The determinant whose elements are exactly the same as those of a square matrix A , is called the **determinant of the square matrix A** and denoted by $|A|$.

Example: If $A = \begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$; then $|A| = \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix}$.

Question: Define singular and non-singular matrix.

Answer: Singular matrix: A square matrix A is said to be a **singular matrix**, if the determinant of A is zero, **i.e.** $|A| = 0$.

Example: Let $A = \begin{bmatrix} 1 & -2 \\ -1 & 2 \end{bmatrix}$; then $|A| = \begin{vmatrix} 1 & -2 \\ -1 & 2 \end{vmatrix} = 0$.

Singular matrix: A square matrix A is said to be a **non-singular matrix**, if the determinant of A is not zero, **i.e.** $|A| \neq 0$.

Example: Let $A = \begin{bmatrix} 1 & 2 \\ -1 & 2 \end{bmatrix}$; then $|A| = \begin{vmatrix} 1 & 2 \\ -1 & 2 \end{vmatrix} \neq 0$.

Question: Define Symmetric and Skew-Symmetric matrix.

Answer: Symmetric matrix: A square matrix $A = [a_{ij}]$ is said to be a **symmetric matrix** if $a_{ij} = a_{ji}$ for all values of i and j **i.e.** $A^t = A$.

Example: Let $A = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$; then $A^t = \begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix} = A$.

Here A is A symmetric matrix.

Skew-Symmetric matrix: A square matrix $A = [a_{ij}]$ is said to be a **skew-symmetric matrix** if $a_{ij} = -a_{ji}$ for all values of i and j **i.e.** $A^t = -A$.

Example: Let $A = \begin{bmatrix} 0 & h & g \\ -h & 0 & f \\ -g & -f & 0 \end{bmatrix}$; then $A^t = \begin{bmatrix} 0 & -h & -g \\ h & 0 & -f \\ g & f & 0 \end{bmatrix} = -\begin{bmatrix} 0 & h & g \\ -h & 0 & f \\ -g & -f & 0 \end{bmatrix} = -A$.

Here A is a skew- symmetric matrix.

NOTE: Every square matrix can be expressed uniquely as the sum of a symmetric matrix and a skew-symmetric matrix.

- ❖ Let A be a square matrix of order n and A^T be the transpose of A . Then we can write,

$$A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

where

$$\frac{1}{2}(A + A^T) = \text{Symmetric matrix.}$$

$$\frac{1}{2}(A - A^T) = \text{Skew-symmetric matrix.}$$

Question: Define Idempotent matrix.

Answer: Idempotent matrix: A square matrix $A = [a_{ij}]_{n \times n}$ is said to be an Idempotent matrix if $A^2 = A$.

Example: $A = \begin{bmatrix} -1 & 3 & 5 \\ 1 & -3 & -5 \\ -1 & 3 & 5 \end{bmatrix}$

Question: Define Involutory matrix.

Answer: Involutory matrix: A square matrix $A = [a_{ij}]_{n \times n}$ is said to be an Involutory matrix if $A^2 = I$.

Example: $A = \begin{bmatrix} 4 & 3 \\ -5 & -4 \end{bmatrix}; B = \begin{bmatrix} 2 & -1 \\ 3 & -2 \end{bmatrix}$

Question: Define an orthogonal matrix.

Answer: Orthogonal matrix: A square matrix $A = [a_{ij}]_{n \times n}$ is said to be an orthogonal matrix if $AA^t = I$, where I is an identity matrix and A^t is the transposed matrix of A .

Example: $A = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}; B = \begin{bmatrix} \frac{1}{3} & \frac{2}{3} & -\frac{2}{3} \\ \frac{2}{3} & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & -\frac{2}{3} & -\frac{1}{3} \end{bmatrix}$

Question: Define Trace or Spur of a matrix.

Answer: Let A be a square matrix of order n . Then the sum of the elements of A lying along the principal diagonal is called the trace of A . We shall write trace of A as $\text{tr}(A)$.

Example: Let $A = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$

So that $\text{tr}(A) = \cos \theta + \cos \theta = 2 \cos \theta$.

Problem-01: If $A = \begin{pmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{pmatrix}$ then find $A+B$ and $2A-B$.

Solution: The given matrices are,

$$A = \begin{pmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{pmatrix}$$

$$\text{Now, } A+B = \begin{pmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{pmatrix} + \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1+3 & 2-1 & -3+2 \\ 5+4 & 0+2 & 2+5 \\ 1+2 & -1+0 & 1+3 \end{pmatrix}$$

$$= \begin{pmatrix} 4 & 1 & -1 \\ 9 & 2 & 7 \\ 3 & -1 & 4 \end{pmatrix}$$

$$\text{Again, } 2A-B = 2 \begin{pmatrix} 1 & 2 & -3 \\ 5 & 0 & 2 \\ 1 & -1 & 1 \end{pmatrix} - \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2 & 4 & -6 \\ 10 & 0 & 4 \\ 2 & -2 & 2 \end{pmatrix} - \begin{pmatrix} 3 & -1 & 2 \\ 4 & 2 & 5 \\ 2 & 0 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 2-3 & 4+1 & -6-2 \\ 10-4 & 0-2 & 4-5 \\ 2-2 & -2-0 & 2-3 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 5 & -8 \\ 6 & -2 & -1 \\ 0 & -2 & -1 \end{pmatrix}.$$

Problem-02: If $A = \begin{pmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{pmatrix}$, $B = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$ and $C = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{pmatrix}$ then find AB and BC .

Solution: The given matrices are,

$$A = \begin{pmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{pmatrix}, B = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \text{ and } C = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{pmatrix}$$

$$\text{Now, } AB = \begin{pmatrix} 1 & 4 & 0 \\ 2 & 5 & 0 \\ 3 & 6 & 0 \end{pmatrix} \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}$$

$$= \begin{pmatrix} 1.3+4.1+0.4 & 1.2+4.2+0.5 & 1.1+4.3+0.6 \\ 2.3+5.1+0.4 & 2.2+5.2+0.5 & 2.1+5.3+0.6 \\ 3.3+6.1+0.4 & 3.2+6.2+0.5 & 3.1+6.3+0.6 \end{pmatrix}$$

$$= \begin{pmatrix} 3+4+0 & 2+8+0 & 1+12+0 \\ 6+5+0 & 4+10+0 & 2+15+0 \\ 9+6+0 & 6+12+0 & 3+18+0 \end{pmatrix}$$

$$= \begin{pmatrix} 7 & 10 & 13 \\ 11 & 14 & 17 \\ 15 & 18 & 21 \end{pmatrix}.$$

$$\text{Again, } BC = \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} 3 & 2 & 1 \\ 1 & 2 & 3 \\ 7 & 8 & 9 \end{pmatrix}$$

$$= \begin{pmatrix} 3.3+2.1+1.7 & 3.2+2.2+1.8 & 3.1+2.3+1.9 \\ 1.3+2.1+3.7 & 1.2+2.2+3.8 & 1.1+2.3+3.9 \\ 4.3+5.1+6.7 & 4.2+5.2+6.8 & 4.1+5.3+6.9 \end{pmatrix}$$

$$= \begin{pmatrix} 9+2+7 & 6+4+8 & 3+6+9 \\ 3+2+21 & 2+4+24 & 1+6+27 \\ 12+5+42 & 8+10+48 & 4+15+54 \end{pmatrix}$$

$$= \begin{pmatrix} 18 & 18 & 18 \\ 26 & 30 & 34 \\ 59 & 66 & 73 \end{pmatrix}.$$

Problem-03: Express $A = \begin{pmatrix} -1 & 2 & 3 \\ 0 & 1 & -2 \\ 4 & -5 & -3 \end{pmatrix}$ as the sum of a Symmetric and Skew-symmetric matrices.

Solution: The given matrix is,

$$A = \begin{pmatrix} -1 & 2 & 3 \\ 0 & 1 & -2 \\ 4 & -5 & -3 \end{pmatrix}$$

The transpose of $\mathbf{A}$ is $A^t = \begin{pmatrix} -1 & 0 & 4 \\ 2 & 1 & -5 \\ 3 & -2 & -3 \end{pmatrix}$

The symmetric matrix of $\mathbf{A}$ is,

$$\frac{1}{2}(A + A^t) = \frac{1}{2} \left\{ \begin{pmatrix} -1 & 2 & 3 \\ 0 & 1 & -2 \\ 4 & -5 & -3 \end{pmatrix} + \begin{pmatrix} -1 & 0 & 4 \\ 2 & 1 & -5 \\ 3 & -2 & -3 \end{pmatrix} \right\}$$

$$= \frac{1}{2} \begin{pmatrix} -2 & 2 & 7 \\ 2 & 2 & -7 \\ 7 & -7 & -6 \end{pmatrix}$$

$$= \begin{pmatrix} -1 & 1 & \frac{7}{2} \\ 1 & 1 & -\frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & -3 \end{pmatrix}$$

The skew-symmetric matrix of $\mathbf{A}$ is,

$$\frac{1}{2}(A - A^t) = \frac{1}{2} \left\{ \begin{pmatrix} -1 & 2 & 3 \\ 0 & 1 & -2 \\ 4 & -5 & -3 \end{pmatrix} - \begin{pmatrix} -1 & 0 & 4 \\ 2 & 1 & -5 \\ 3 & -2 & -3 \end{pmatrix} \right\}$$

$$= \frac{1}{2} \begin{pmatrix} 0 & 2 & -1 \\ -2 & 0 & 3 \\ 1 & -3 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & 1 & -\frac{1}{2} \\ -1 & 0 & \frac{3}{2} \\ \frac{1}{2} & -\frac{3}{2} & 0 \end{pmatrix}$$

$$\text{Therefore, } A = \begin{pmatrix} -1 & 1 & \frac{7}{2} \\ 1 & 1 & -\frac{7}{2} \\ \frac{7}{2} & -\frac{7}{2} & -3 \end{pmatrix} + \begin{pmatrix} 0 & 1 & -\frac{1}{2} \\ -1 & 0 & \frac{3}{2} \\ \frac{1}{2} & -\frac{3}{2} & 0 \end{pmatrix}.$$

Problem-04: Express $A = \begin{pmatrix} 1 & 0 & 6 \\ 4 & 4 & -1 \\ 0 & -9 & 6 \end{pmatrix}$ as the sum of a Symmetric and Skew-symmetric matrices.

Solution: The given matrix is,

$$A = \begin{pmatrix} 1 & 0 & 6 \\ 4 & 4 & -1 \\ 0 & -9 & 6 \end{pmatrix}$$

$$\text{The transpose of } \mathbf{A} \text{ is } A^t = \begin{pmatrix} 1 & 4 & 0 \\ 0 & 4 & -9 \\ 6 & -1 & 6 \end{pmatrix}$$

The symmetric matrix of $\mathbf{A}$ is,

$$\frac{1}{2}(A + A^t) = \frac{1}{2} \left\{ \begin{pmatrix} 1 & 0 & 6 \\ 4 & 4 & -1 \\ 0 & -9 & 6 \end{pmatrix} + \begin{pmatrix} 1 & 4 & 0 \\ 0 & 4 & -9 \\ 6 & -1 & 6 \end{pmatrix} \right\}$$

$$= \frac{1}{2} \begin{pmatrix} 2 & 4 & 6 \\ 4 & 8 & -10 \\ 6 & -10 & 12 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & -5 \\ 3 & -5 & -6 \end{pmatrix}$$

The skew-symmetric matrix of A is,

$$\frac{1}{2}(A - A^t) = \frac{1}{2} \left\{ \begin{pmatrix} 1 & 0 & 6 \\ 4 & 4 & -1 \\ 0 & -9 & 6 \end{pmatrix} - \begin{pmatrix} 1 & 4 & 0 \\ 0 & 4 & -9 \\ 6 & -1 & 6 \end{pmatrix} \right\}$$

$$= \frac{1}{2} \begin{pmatrix} 0 & -4 & 6 \\ 4 & 0 & 8 \\ -6 & -8 & 0 \end{pmatrix}$$

$$= \begin{pmatrix} 0 & -2 & 3 \\ 2 & 0 & 4 \\ -3 & -4 & 0 \end{pmatrix}$$

Therefore, $A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & -5 \\ 3 & -5 & 6 \end{pmatrix} + \begin{pmatrix} 0 & -2 & 3 \\ 2 & 0 & 4 \\ -3 & -4 & 0 \end{pmatrix}.$

Exercise: 1. Find the Symmetric and Skew-symmetric parts of the matrix

$$A = \begin{pmatrix} 1 & 2 & 4 \\ 6 & 8 & 1 \\ 3 & 5 & 7 \end{pmatrix}.$$

2. Find the Symmetric and Skew-symmetric parts of the matrix

$$A = \begin{pmatrix} 1 & \frac{1}{3} & 1 & 4 \\ 1 & -1 & 0 & -1 \\ -3 & 0 & -\frac{2}{5} & 6 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$